

Chapter 11: Fluid Statics

T. K. Deskins, Ph.D.

Professor of Physics & Chemistry
Southern Union State Community College

PHY201
Lecture 11



Part I

What is a Fluid?

Outline of Part I: What is a Fluid?

What is a Fluid?

Density

What is a Fluid?

Definition: Fluid

A **fluid** is any substance that *yields to shearing (sideways) forces* — it flows and takes the shape of its container.

Fluids

- ▶ Liquids (water, mercury, oil)
- ▶ Gases (air, steam, helium)
- ▶ Plasmas

Not Fluids

- ▶ Solids (steel, wood, bone)
- ▶ Solids *resist* shearing forces and maintain their shape

Fluid Statics is the study of *fluids at rest*.

When a fluid is at rest, it exerts forces only *perpendicular* to any surface — never parallel.

States of Matter Compared

Phase	Shape	Volume	Compressible?
Solid	Definite	Definite	No (atoms in fixed lattice)
Liquid	Takes container	Definite	No (atoms closely packed)
Gas	Takes container	Variable	Yes (atoms widely spaced)
Plasma	Takes container	Variable	Yes (free ions/electrons)

Liquids and solids have similar densities because their atoms are in contact. Gases are $\sim 1000\times$ less dense — atoms are far apart and weakly interacting.

Check Your Understanding

Which of the following are fluids at room temperature?

Air Mercury Water Glass

What physical characteristic distinguishes a fluid from a solid?

Come to lecture to see the answer.

Outline of Part I: What is a Fluid?

What is a Fluid?

Density

Why Density? Why Not Mass?

For solids, we often use **mass** to quantify “how much stuff” there is.

But fluids *spread to fill any container*, so their mass alone says little about how they behave.

∴ For fluids, **volume** is an accessible and natural measurement.

We define a quantity that relates mass and volume:

Density ρ — SI Unit: $\frac{\text{kg}}{\text{m}^3}$

$$\rho = \frac{m}{V}$$

where m is mass (kg) and V is volume (m^3).

Useful conversions: $1000 \text{ L} = 1 \text{ m}^3$; $1 \text{ cc} = 1 \text{ cm}^3 = 1 \text{ mL}$

Note: $1 \frac{\text{g}}{\text{cm}^3} = 1 \frac{\text{g}}{\text{mL}} = 1000 \frac{\text{kg}}{\text{m}^3}$

Selected Densities

Substance	ρ (kg/m ³)
Gold	19,320
Mercury	13,600
Lead	11,300
Iron/Steel	7,800
Aluminum	2,700
Bone	1,900

Substance	ρ (kg/m ³)
Water (4°C)	1,000
Blood (37°C)	1,050
Ice	917
Ethyl alcohol	790
Air (15°C)	1.225
Helium	0.164

Object floats in a fluid when $\rho_{\text{obj}} < \rho_{\text{fluid}}$; sinks when $\rho_{\text{obj}} > \rho_{\text{fluid}}$.

Ice floats on water because $\rho_{\text{ice}} = 917 \frac{\text{kg}}{\text{m}^3} < \rho_{\text{water}}$.

Example: Mass of Water in a Reservoir

A reservoir has a surface area of 50.0 km^2 and an average depth of 40.0 m . What is the mass of water stored in the reservoir?

$$V = A \times h$$

$$V = (50.0 \text{ km}^2) \left(\frac{10^3 \text{ m}}{1 \text{ km}} \right)^2 \times 40.0 \text{ m}$$

$$V = 2.00 \times 10^9 \text{ m}^3$$

$$m = \rho V = \left(1.00 \times 10^3 \frac{\text{kg}}{\text{m}^3} \right) (2.00 \times 10^9 \text{ m}^3)$$

$$m = 2.00 \times 10^{12} \text{ kg}$$

- ▶ Convert km^2 to m^2 carefully:
 $1 \text{ km}^2 = (10^3 \text{ m})^2 = 10^6 \text{ m}^2$
- ▶ $\rho_{\text{water}} = 1000 \frac{\text{kg}}{\text{m}^3}$
- ▶ $2 \times 10^{12} \text{ kg} = 2 \text{ trillion kg} \approx 2 \text{ gigatons}$

Check Your Understanding

A glass of ice water is filled exactly to the brim.
Will water overflow when the ice melts? Explain using density.

Come to lecture to see the answer.

Part II

Pressure

Outline of Part II: Pressure

Pressure

Pressure Variation with Depth

Pressure

Definition: Pressure P — SI Unit: Pascal (Pa)

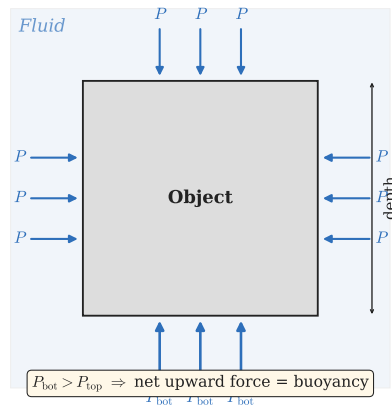
Pressure is the *perpendicular* force applied per unit area:

$$P = \frac{F}{A}$$

where F is the force (N) and A is the area (m^2).

$$1 \text{ Pa} = 1 \frac{\text{N}}{\text{m}^2}$$

- ▶ A static fluid exerts force *only perpendicular* to any surface — never parallel (no shear).
- ▶ Pressure is *isotropic*: at a given depth, pressure is the same in all directions.



Common Pressure Units

- ▶ $1 \text{ atm} = 101\,325 \text{ Pa} = 760 \text{ mmHg}$
- ▶ $1 \text{ atm} = 14.7 \text{ psi}$

Pressure and Area: Everyday Examples

- ▶ A **sharp knife** exerts far more pressure than a dull one for the same applied force — because the blade edge has a much smaller area A .
- ▶ A **sharp hypodermic needle** hurts less than a dull one: smaller $A \Rightarrow$ cuts through tissue with less tearing.
- ▶ A dancer on **toe shoes** puts enormous pressure on the toes (A is very small): same weight, tiny area.
- ▶ **Snowshoes** spread your weight over a large A , reducing P and preventing you from sinking into snow.

Same force, smaller area $\Rightarrow$ larger pressure. $P = F/A$

Check Your Understanding

A 60.0 kg person stands on one foot.

The area of their foot in contact with the floor is 150 cm^2 .

What gauge pressure do they exert on the floor?

Express your answer in Pa and in atm.

Come to lecture to see the answer.

Outline of Part II: Pressure

Pressure

Pressure Variation with Depth

Pressure Increases with Depth

Pressure at Depth h in a Fluid

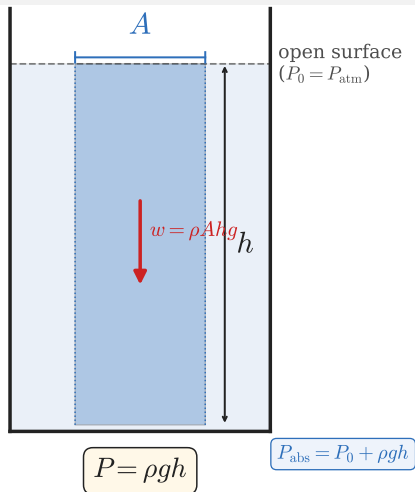
Consider a fluid column of height h , density ρ , cross-section A :

$$P = \frac{F}{A} = \frac{mg}{A} = \frac{\rho Vg}{A} = \frac{\rho(Ah)g}{A}$$

$$P = \rho gh$$

The **absolute pressure** at depth h below an open surface:

$$P = P_{\text{atm}} + \rho gh$$



- P depends only on depth h and density ρ — *not* on the container shape.

Isotropic Pressure in a Fluid

Consider any point inside a static fluid at depth h below the surface:

$$P_{\text{red dot}} = P_0 + \rho gh$$

where P_0 is the pressure at the surface (often P_{atm}).

The force on a horizontal surface of area $A = LW$ at that depth:

$$F = P \cdot A = P \cdot LW$$

Key insight: All points at the same depth in a connected fluid experience the same pressure, regardless of horizontal position. This is why the pressure at the bottom of a lake is determined only by the depth, not the shape of the lake.

Example: Depth of Water Equivalent to 1 atm

At what depth below the surface of water is the gauge pressure equal to $P_{\text{atm}} = 101\,325\text{ Pa}$?

$$P = \rho gh \quad \Rightarrow \quad h = \frac{P}{\rho g}$$

$$h = \frac{101\,325\text{ Pa}}{(1000\frac{\text{kg}}{\text{m}^3})(9.80\frac{\text{m}}{\text{s}^2})}$$

$$h = 10.3\text{ m}$$

Just 10.3 m of water produces the same pressure as the entire 120-km-thick atmosphere.

Water is $\sim 775\times$ denser than air.

Example: Force on a Dam

A dam is 500 m wide and holds water to a depth of 80.0 m.
Find (a) the average pressure on the dam face and (b) the total force on the dam.

$$(a) \quad \bar{h} = \frac{80.0 \text{ m}}{2} = 40.0 \text{ m}$$

$$\bar{P} = \rho g \bar{h} = (1000)(9.80)(40.0)$$

$$\boxed{\bar{P} = 3.92 \times 10^5 \text{ Pa} = 392 \text{ kPa}}$$

$$(b) \quad A_{\text{dam}} = (80.0)(500) = 4.00 \times 10^4 \text{ m}^2$$

$$F = \bar{P} \cdot A = (3.92 \times 10^5)(4.00 \times 10^4)$$

$$\boxed{F = 1.57 \times 10^{10} \text{ N}}$$

Pressure increases linearly with depth, so the average pressure occurs at the average depth $\bar{h} = h/2$.

$1.57 \times 10^{10} \text{ N} \approx 1.6$ billion kg-weight of force. This is why dam construction is one of the most demanding engineering tasks.

Check Your Understanding

Atmospheric pressure is about $101\,325\text{ Pa}$ — roughly the equivalent of 10 metric tons pressing on every square meter of your body.

Why aren't you crushed?

Come to lecture to see the answer.

Check Your Understanding

A scuba diver descends to 30.0 m in seawater ($\rho = 1025 \frac{\text{kg}}{\text{m}^3}$).

What is the absolute pressure at this depth? Express in Pa and in atm.

Come to lecture to see the answer.

Part III

Pascal's Principle & Gauge Pressure

Outline of Part III: Pascal's Principle & Gauge Pressure

Pascal's Principle

Gauge Pressure and Absolute Pressure

Pascal's Principle

Pascal's Principle

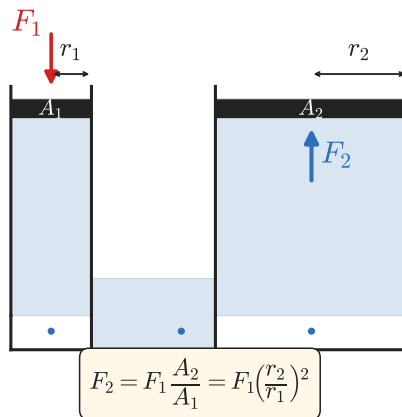
A change in pressure applied to an *enclosed* fluid is transmitted **undiminished** to all portions of the fluid and the walls of its container.

Hydraulic System: Two Pistons

For two pistons connected by enclosed fluid (at the same height):

$$P_1 = P_2 \quad \Rightarrow \quad \frac{F_1}{A_1} = \frac{F_2}{A_2}$$

$$\therefore F_2 = F_1 \frac{A_2}{A_1} = F_1 \left(\frac{r_2}{r_1} \right)^2$$



If $r_1 < r_2$, then $F_2 > F_1$:

$\Rightarrow$ a small force on a small piston produces a large force on a large piston.

Force Amplification Without Energy Creation

Energy is still conserved. A hydraulic system amplifies *force*, not energy. The large piston moves a shorter distance than the small piston:

$$W_{\text{in}} = W_{\text{out}} \quad \Rightarrow \quad F_1 d_1 = F_2 d_2$$

Applications of Pascal's Principle

- ▶ **Car brakes:** small pedal force $\Rightarrow$ large braking force at each wheel
- ▶ **Hydraulic lifts:** garage lifts, excavators, forklifts
- ▶ **Power steering:** small hand force $\Rightarrow$ large force to turn tires
- ▶ **Hydraulic press:** crushing, forming, stamping metal

Example: Car Brake System

A hydraulic brake has a pedal cylinder of diameter 0.500 cm and wheel cylinders of diameter 2.50 cm. The driver applies $F_1 = 500$ N. Find the braking force F_2 at each wheel cylinder.

$$F_2 = F_1 \left(\frac{r_2}{r_1} \right)^2$$

$$F_2 = 500 \text{ N} \left(\frac{1.25 \text{ cm}}{0.250 \text{ cm}} \right)^2$$

$$F_2 = 500 \text{ N} \times 25$$

$$F_2 = 1.25 \times 10^4 \text{ N per wheel}$$

A 500 N foot force (≈ 112 lbs) produces 12 500 N (≈ 2800 lbs) at each wheel.

The 25 $\times$ amplification comes from the area ratio: $(r_2/r_1)^2 = 5^2 = 25$.

Check Your Understanding

A hydraulic lift has input radius $r_1 = 5.0$ cm and output radius $r_2 = 15.0$ cm.

- (a) By what factor is the force amplified?
- (b) If the input piston moves down 9.0 cm, how far does the output piston rise?

Come to lecture to see the answer.

Outline of Part III: Pascal's Principle & Gauge Pressure

Pascal's Principle

Gauge Pressure and Absolute Pressure

Gauge Pressure vs. Absolute Pressure

Two Ways to Measure Pressure

$$P_{\text{abs}} = P_{\text{gauge}} + P_{\text{atm}}$$

Absolute Pressure P_{abs}

- ▶ Total pressure, including atmosphere
- ▶ Zero only in a perfect vacuum
- ▶ $P_{\text{abs}} = P_{\text{atm}}$ at an open surface

Gauge Pressure P_g

- ▶ Pressure *relative to* atmosphere
- ▶ $P_g = 0$ when $P_{\text{abs}} = P_{\text{atm}}$
- ▶ $P_g < 0$ in partial vacuum (shop-vac)
- ▶ From a fluid column: $P_g = \rho gh$

Most gauges (tire gauges, blood pressure cuffs) read *gauge* pressure. A tire gauge showing zero does not mean a vacuum — the tire is simply at atmospheric pressure.

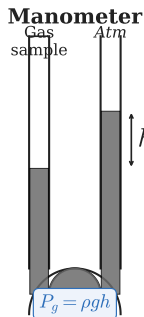
Pressure Measurement Devices

Open-Tube Manometer

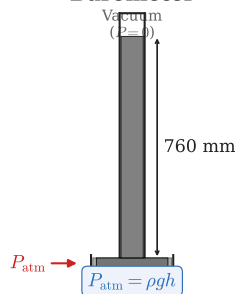
- ▶ U-tube with one end open to atmosphere, other connected to gas sample
- ▶ Height difference h gives gauge pressure:

$$P_g = \rho gh$$

- ▶ Both sides equal $\Rightarrow$
 $P_g = 0$ (gas at atmospheric pressure)



Barometer



Mercury Barometer

- ▶ Evacuated tube inverted in mercury dish
- ▶ Column Pressure:
 $P_{\text{abs}} \approx 760 \text{ mm} = 1 \text{ atm}$
- ▶ Mercury used because it is dense: column stays at manageable height ($\sim 760 \text{ mm}$ vs. $\sim 10.3 \text{ m}$ for water)

Blood Pressure

Blood pressure is reported as gauge pressure in mmHg, measured at heart level.

Normal: **120/80 mm Hg** (systolic / diastolic)

- ▶ **Systolic:** peak pressure during heart contraction (~ 120 mm Hg)
- ▶ **Diastolic:** minimum pressure between beats (~ 80 mm Hg)

Mercury manometers are preferred for blood pressure because $\rho_{\text{Hg}} = 13\,600 \frac{\text{kg}}{\text{m}^3}$, so the column stays at ~ 120 mm.

A water manometer would need $120 \text{ mmHg} \times 13.6 \approx 1.6$ m of column height — impractical for clinical use.

Check Your Understanding

An IV bag must deliver fluid into a vein with gauge pressure 18.0 mmHg.

How high above the vein must the bag be held?

($\rho_{\text{fluid}} = 1000 \frac{\text{kg}}{\text{m}^3}$; 1 mmHg = 133 Pa)

Come to lecture to see the answer.

Part IV

Archimedes' Principle

Outline of Part IV: Archimedes' Principle

Archimedes' Principle

Buoyancy: The Physical Origin

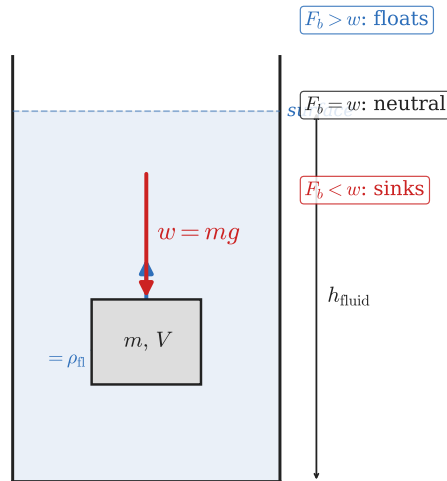
$\therefore$ pressure increases with depth, the upward force on the *bottom* of a submerged object is greater than the downward force on its *top*. This net upward force is the **buoyant force** F_b .

Archimedes' Principle

The buoyant force equals the *weight of the fluid displaced*:

$$F_b = w_{\text{fl}} = \rho_{\text{fl}} V_{\text{sub}} g$$

where V_{sub} is the volume of fluid displaced (= submerged volume of the object).



$$\rho_{\text{obj}} < \rho_{\text{fl}}: \text{floats}$$

$$\rho_{\text{obj}} = \rho_{\text{fl}}: \text{neutral buoyancy}$$

$$\rho_{\text{obj}} > \rho_{\text{fl}}: \text{sinks}$$

Fraction of Object Submerged

Floating Equilibrium: $F_b = w_{\text{obj}}$

$$\rho_{\text{fl}} V_{\text{sub}} g = \rho_{\text{obj}} V_{\text{obj}} g \quad \Rightarrow \quad f_{\text{sub}} = \frac{V_{\text{sub}}}{V_{\text{obj}}} = \frac{\rho_{\text{obj}}}{\rho_{\text{fl}}}$$

Example: Iceberg

$$f_{\text{sub}} = \frac{\rho_{\text{ice}}}{\rho_{\text{water}}} = \frac{917 \frac{\text{kg}}{\text{m}^3}}{1000 \frac{\text{kg}}{\text{m}^3}} = 0.917$$

$\therefore$ 91.7% of an iceberg is below the waterline. Only $\sim 8\%$ is visible above — which is why the phrase “tip of the iceberg” describes a hidden majority.

Example: Steel Sinks, Steel Ship Floats

A solid steel block ($m = 1.00 \times 10^7 \text{ kg}$, $\rho_{\text{steel}} = 7800 \frac{\text{kg}}{\text{m}^3}$) is placed in water.

(a) Does the solid block sink?

(b) Shaped into a hollow hull displacing $V = 1.00 \times 10^5 \text{ m}^3$, can it float?

$$(a) \quad V_{\text{steel}} = \frac{m}{\rho_{\text{steel}}} = \frac{1.00 \times 10^7}{7800} = 1282 \text{ m}^3$$

$$F_b = \rho_w V_{\text{steel}} g = (1000)(1282)(9.80)$$

$$F_b = 1.26 \times 10^7 \text{ N}$$

$$w = mg = 9.80 \times 10^7 \text{ N} \gg F_b \Rightarrow \text{ **sinks** }$$

$$(b) \quad F_b = \rho_w V_{\text{hull}} g$$

$$F_b = (1000)(10^5)(9.80)$$

$$F_b = 9.80 \times 10^8 \text{ N}$$

$$F_b \gg w \Rightarrow \text{ **floats!** }$$

Shaping steel into a hollow hull increases the displaced volume enormously without increasing the mass.

Archimedes' Density Test

Apparent Weight in a Fluid

A submerged object appears lighter because F_b partially supports it:

$$w_{\text{apparent}} = w_{\text{air}} - F_b$$

The “lost weight” = weight of fluid displaced. This allows density measurement without separately measuring volume.

Example: Authenticating a Coin

Coin: $m_{\text{air}} = 8.630$ g; apparent mass submerged in water: 7.800 g.

$$m_{\text{displaced}} = 8.630 - 7.800 = 0.830 \text{ g} \Rightarrow V_{\text{coin}} = 0.830 \text{ cm}^3$$

$$\rho_{\text{coin}} = \frac{8.630}{0.830} = 10.4 \frac{\text{g}}{\text{cm}^3}$$

$$\rho_{\text{silver}} = 10.5 \frac{\text{g}}{\text{cm}^3} \checkmark \quad (\rho_{\text{gold}} = 19.3 \frac{\text{g}}{\text{cm}^3} \text{ — not gold!})$$

Check Your Understanding

An object has mass 4.00 kg in air. When fully submerged in water, it appears to have mass 3.20 kg.

- (a) What is the buoyant force?
- (b) What is the volume of the object?
- (c) What is the density of the object?

Come to lecture to see the answer.

Check Your Understanding

Do fluids exert buoyant forces in a weightless environment (e.g., the International Space Station)?

Use the physical origin of buoyancy to explain your answer.

Come to lecture to see the answer.

Part V

Cohesion, Adhesion & Surface Tension

Outline of Part V: Cohesion, Adhesion & Surface Tension

Cohesion and Adhesion

Cohesion and Adhesion

Cohesive Forces

Attractive forces between molecules of the **same** substance.

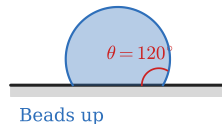
Example: Water–water hydrogen bonds (strong cohesion).

Adhesive Forces

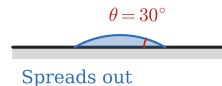
Attractive forces between molecules of **different** substances.

Example: Water–glass attraction (strong adhesion).

Water on Wax
Cohesion > Adhesion



Water on Glass
Adhesion > Cohesion



The **contact angle** θ shows which force dominates:

- ▶ $\theta < 90^\circ$: adhesion > cohesion (liquid wets surface, spreads)
- ▶ $\theta > 90^\circ$: cohesion > adhesion (liquid beads up)

Surface Tension

Surface Tension γ — SI Unit: N/m

Molecules at a liquid surface are pulled *inward* by cohesive forces, with no outward neighbor to balance them. The surface contracts to the minimum area, behaving like a stretched elastic film.

$$\gamma = \frac{F}{L}$$

where F is the force along a surface line of length L .

Selected Values (at 20°C)

Substance	γ (N/m)
Mercury	0.465
Blood (37°C)	0.058
Water	0.0728
Soapy water	0.0370

Surface tension explains:

- ▶ Water striders walking on water
- ▶ Soap bubbles and droplets forming spheres (minimum surface area)
- ▶ Lung alveoli staying open
- ▶ Needle floating on still water

Pressure Inside a Bubble

Gauge Pressure Inside a Spherical Soap Bubble

A soap bubble has **two** surfaces (inner and outer wall of the soap film), each contributing. The gauge pressure inside:

$$P_g = \frac{4\gamma}{r}$$

For a single curved surface (e.g. one side of a droplet): $P_g = 2\gamma/r$.

Example: Pressure Inside a Soap Bubble

Radius $r = 2.00 \times 10^{-4} \text{ m}$; $\gamma_{\text{soap}} = 0.0370 \frac{\text{N}}{\text{m}}$:

$$P_g = \frac{4\gamma}{r} = \frac{4(0.0370)}{2.00 \times 10^{-4}} = 740 \text{ Pa} = 5.56 \text{ mmHg}$$

Smaller bubbles have *higher* internal pressure. When two connected bubbles differ in size, the smaller bubble collapses into the larger one.

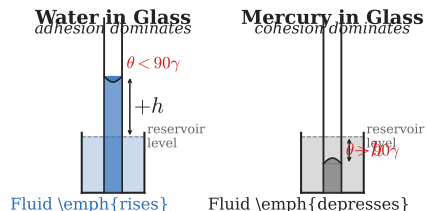
Capillary Action

Capillary Rise/Depression Height

In a tube of radius r , a fluid rises or is depressed by:

$$h = \frac{2\gamma \cos \theta}{\rho g r}$$

- ▶ $\theta < 90^\circ \Rightarrow h > 0$: fluid *rises*
(*water in glass*: $\theta \approx 0^\circ$, concave meniscus)
- ▶ $\theta > 90^\circ \Rightarrow h < 0$: fluid is *depressed*
(*mercury in glass*: $\theta = 140^\circ$, convex meniscus)



Capillary action requires **both**:

- ▶ Adhesion: pulls fluid up walls
- ▶ Cohesion: pulls column along

Smaller $r \Rightarrow$ larger h (the column rises higher in a narrower tube).

Example: Capillary Rise of Water in Glass

Water ($\gamma = 0.0728 \frac{\text{N}}{\text{m}}$, $\theta \approx 0^\circ$) in a glass capillary tube of radius $r = 0.100 \text{ mm}$. How high does the water rise?

$$h = \frac{2\gamma \cos \theta}{\rho g r}$$

$$h = \frac{2(0.0728)(1)}{(1000)(9.80)(1.00 \times 10^{-4})}$$

$$h = 0.149 \text{ m} = 14.9 \text{ cm}$$

$h \propto 1/r$: narrower tube $\Rightarrow$ higher rise.

This is why paper towels, wicks, and sponges are effective absorbers — their fine porous structure provides many narrow capillary channels.

Check Your Understanding

- (a) Is surface tension due to cohesive forces, adhesive forces, or both?
- (b) Is capillary action due to cohesive forces, adhesive forces, or both?
- (c) Could capillary action move fluids in a weightless environment aboard the ISS? Explain.

Come to lecture to see the answer.

Surface Tension in Biology

Where Surface Tension and Capillary Action Matter

- ▶ **Trees:** Water rises through xylem vessels ($r \sim 25 \mu\text{m}$) by capillary action and osmotic pressure. Capillary action alone cannot raise water 100 m — other mechanisms (transpiration pull, osmotic pressure) must contribute.
- ▶ **Lung alveoli:** Surfactant molecules coat alveolar surfaces. Without surfactant, $P_g = 4\gamma/r$ would be enormous in small alveoli (r small), causing them to collapse. Surfactant reduces γ in proportion to surface area.
- ▶ **Water striders:** Insect feet depress the water surface without breaking it because $F_b = \gamma L$ supports the insect's weight.
- ▶ **Manometer readings:** Meniscus curves slightly bias readings; read at the bottom of a concave (water) meniscus.

Part VI

Pressures in the Body

Outline of Part VI: Pressures in the Body

Pressures in the Body

Fluid Pressures in the Human Body

Location	Pressure	Notes
Aorta (systolic / diastolic)	120 / 80 mm Hg	Gauge, at heart level
Foot arteries (standing)	~ 228 mm Hg systolic	$+\rho gh$ for blood column
Intraocular pressure	12–24 mm Hg	Glaucoma: up to 85 mm Hg
Cerebrospinal fluid	5–12 mm Hg	Buoys the brain
Bladder	0–25 mm Hg	Up to 100 mm Hg during voiding
Femur (load-bearing)	$\sim 5 \times 10^6$ Pa	Highest pressure in the body

Blood pressure is always measured at *heart level*. Raising the arm lowers the reading; lowering the arm increases it — by $\Delta P = \rho g \Delta h$.

The Brain Floats: Archimedes in the Body

Cerebrospinal Fluid (CSF)

The brain ($m \approx 1.5 \text{ kg}$) is suspended in cerebrospinal fluid inside the skull:

- ▶ $\rho_{\text{brain}} \approx 1040 \frac{\text{kg}}{\text{m}^3}$; $\rho_{\text{CSF}} \approx 1007 \frac{\text{kg}}{\text{m}^3}$
- ▶ Buoyant force reduces brain's apparent weight from $\sim 15 \text{ N}$ (actual) to $\sim 0.5 \text{ N}$ (apparent) — only about 1/30 of its true weight
- ▶ This protects the brainstem from being crushed by the weight of the cerebrum
- ▶ Rapid deceleration (concussion) causes the brain to move within the skull despite the CSF cushion

This is Archimedes' principle applied inside the skull: the buoyant force of CSF nearly equals the weight of the brain.

Example: Eardrum Rupture Pressure

An eardrum has area $A = 1.00 \text{ cm}^2$ and ruptures under a net force $F = 3.00 \text{ N}$.

(a) What gauge pressure causes rupture?

(b) At what depth in freshwater would this pressure be reached?

$$(a) \quad P_g = \frac{F}{A} = \frac{3.00 \text{ N}}{1.00 \times 10^{-4} \text{ m}^2}$$

$$P_g = 3.00 \times 10^4 \text{ Pa} = 226 \text{ mmHg}$$

$$(b) \quad h = \frac{P_g}{\rho g} = \frac{3.00 \times 10^4 \text{ Pa}}{(1000)(9.80)}$$

$$h = 3.06 \text{ m}$$

Just 3 meters of water can rupture an eardrum if the diver does not equalize pressure by swallowing or yawning (raising the pressure behind the eardrum to match the outside).

Intraocular Pressure and Glaucoma

The eye is filled with **aqueous humor** — fluid that maintains the eyeball's shape. It is continuously produced and drained through a trabecular meshwork.

Normal Eye

- ▶ Intraocular pressure: 12–24 mm Hg gauge
- ▶ Measured by a **tonometer** (the “puff-of-air” test at the eye doctor)
- ▶ Maintains the spherical shape of the eye for focusing

Glaucoma

- ▶ Blocked drainage $\Rightarrow$ pressure rises to ~ 85 mm Hg
- ▶ Optic nerve is compressed $\Rightarrow$ permanent vision loss
- ▶ Treated with eye drops (reduce fluid production) or surgery (improve drainage)

Direct application of $P = F/A$: elevated fluid pressure transmits via Pascal's principle to compress and damage the optic nerve.

Check Your Understanding

A patient's blood pressure is measured at heart level as 120/80 mm Hg.

The cuff is then placed on the ankle, which is 1.20 m *below* the heart. What would the systolic reading be?

($\rho_{\text{blood}} = 1050 \frac{\text{kg}}{\text{m}^3}$; 1 mmHg = 133 Pa)

Come to lecture to see the answer.

Chapter Summary

Key Equations — Chapter 11

Concept	Equation	SI Unit
Density	$\rho = m/V$	kg/m ³
Pressure	$P = F/A$	Pa
Pressure at depth	$P = P_{\text{atm}} + \rho gh$	Pa
Gauge pressure	$P_{\text{abs}} = P_g + P_{\text{atm}}$	Pa
Pascal's Principle	$F_1/A_1 = F_2/A_2$	—
Buoyant force	$F_b = \rho_{\text{fl}} V_{\text{sub}} g$	N
Fraction submerged	$f_{\text{sub}} = \rho_{\text{obj}}/\rho_{\text{fl}}$	—
Surface tension	$\gamma = F/L$	N/m
Bubble pressure	$P_g = 4\gamma/r$	Pa
Capillary height	$h = 2\gamma \cos \theta /(\rho gr)$	m